Introduction to Theoretical Computer Science

CS2490 - Session 7: From NFA to DFA

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Fall 2026

August 31, 2026

Recall

Let Σ be an alphabet. A nondeterministic finite automaton (NFA) $\mathcal{A}$ is given by:

- finite set Q: each element is a name of a state,
- a subset $I \subseteq Q$: each element is a name of an initial state,
- a subset $F \subseteq Q$: each element is a name of an accepting/final state,
- a function $\Delta : \Sigma \times Q \to \mathcal{P}(Q)$ (called the transition relation): when in state q, and reading a, change to one of the states in $\Delta(a, q)$.

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Two sources of nondeterminism:

- Nowhere to go
- Too many places to go

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Potentially many computations for a given word. Or none at all!

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The language accepted by $\mathcal{A}$ is the set of all words with at least one successful computation:

$$L(\mathcal{A}) = \{w \in \Sigma^* \mid \bigcup_{i \in I} \Delta^*(w, i) \cap F \neq \emptyset\}$$

Why NFAs?

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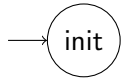
This is a trade-off: NFA are easier to design, but harder to run.

Practice

Make a NFA over $\{a, b\}$ that accepts words with at least two as.

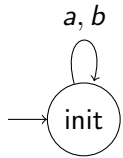
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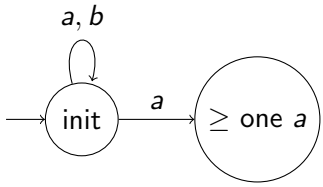
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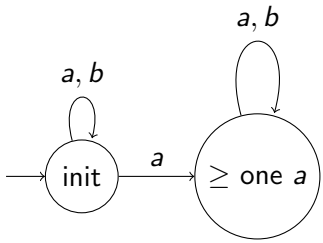
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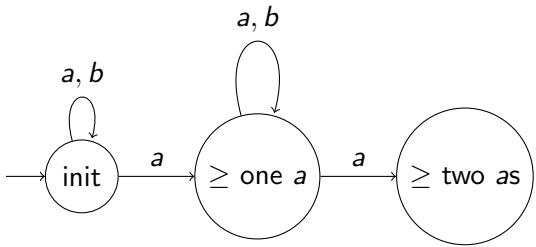
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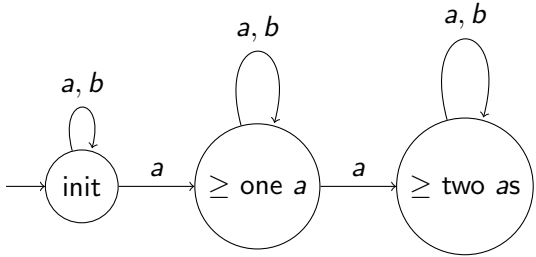
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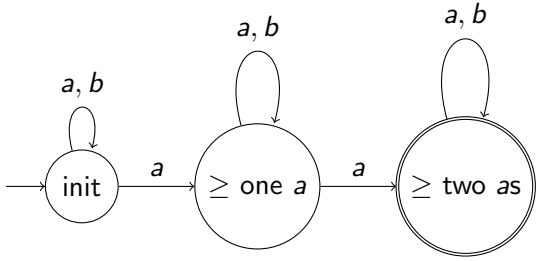
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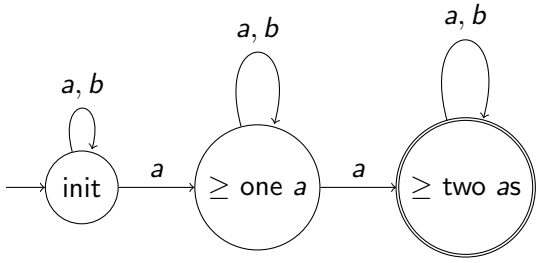
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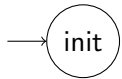
How many runs for *abbaaababababa*?

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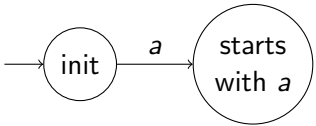
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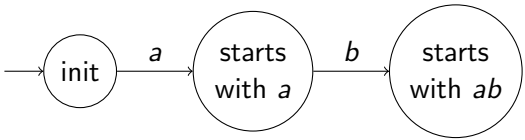
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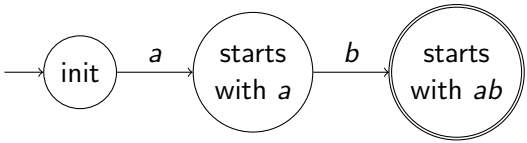
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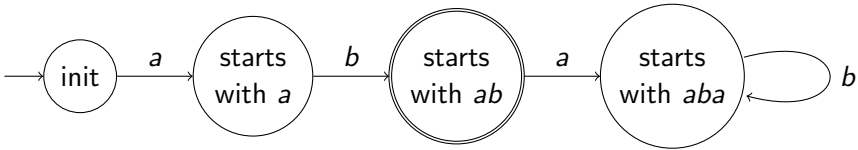
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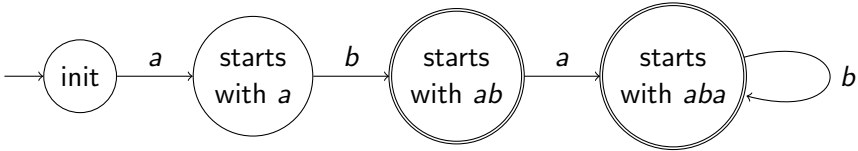
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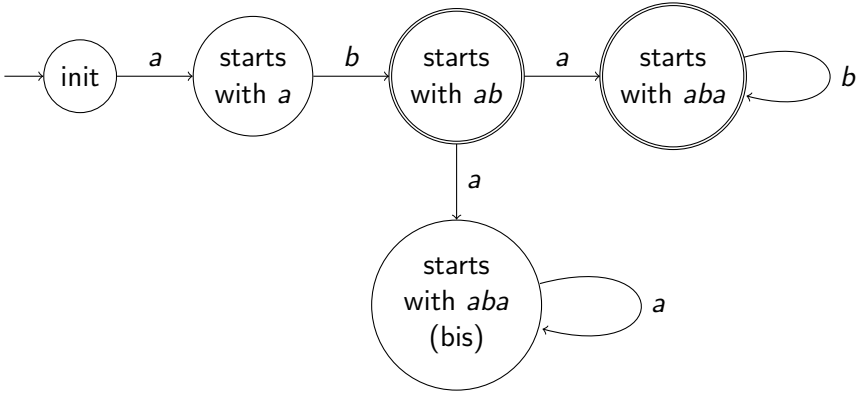
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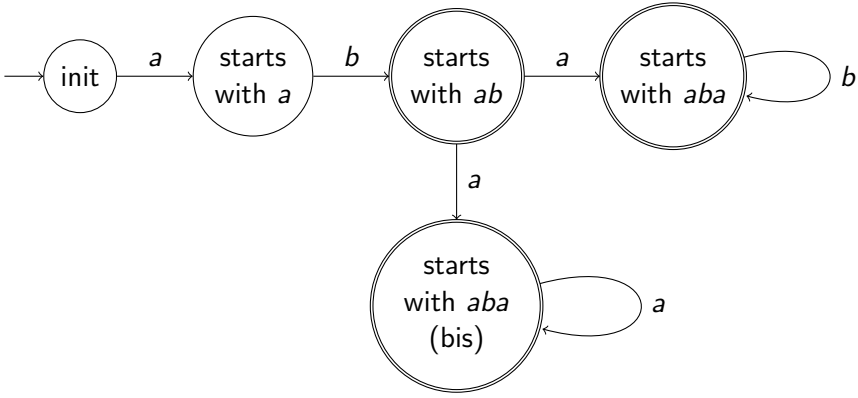
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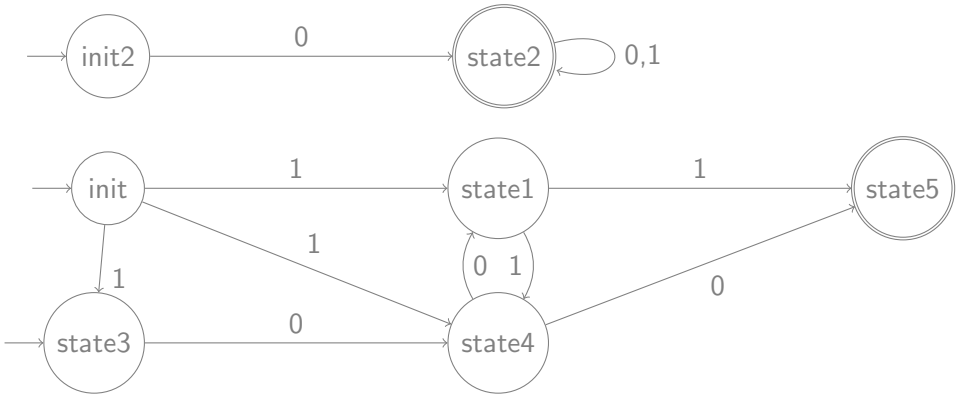
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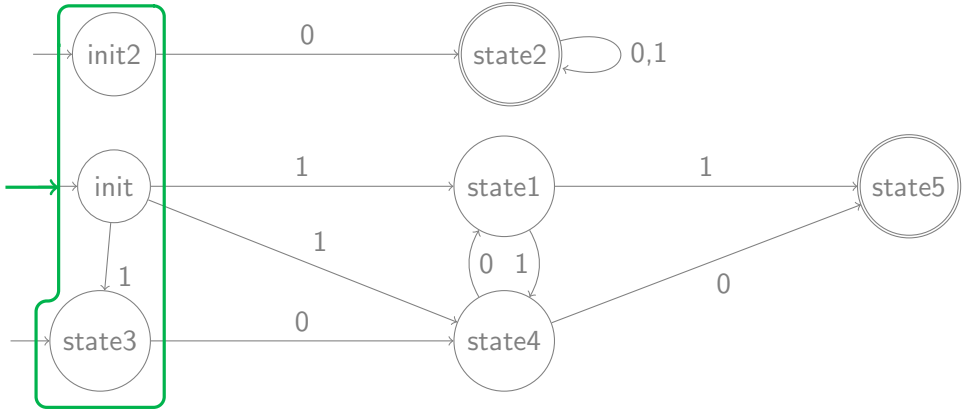
Can we transform an NFA into an *equivalent* DFA?

Need to transform $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ into $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$ such that $\mathrm{L}(\mathcal{A}) = \mathrm{L}(\mathcal{A}^D)$.

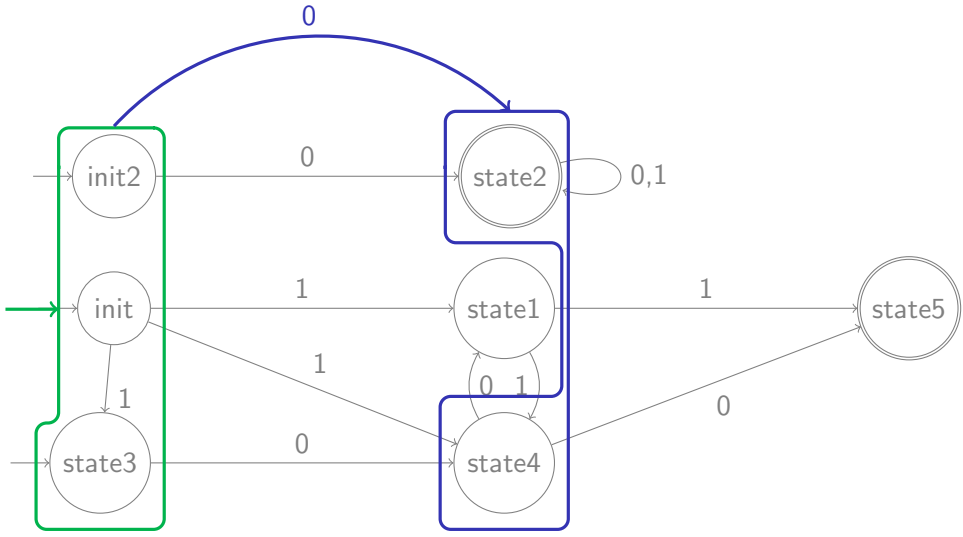
Trial by fire



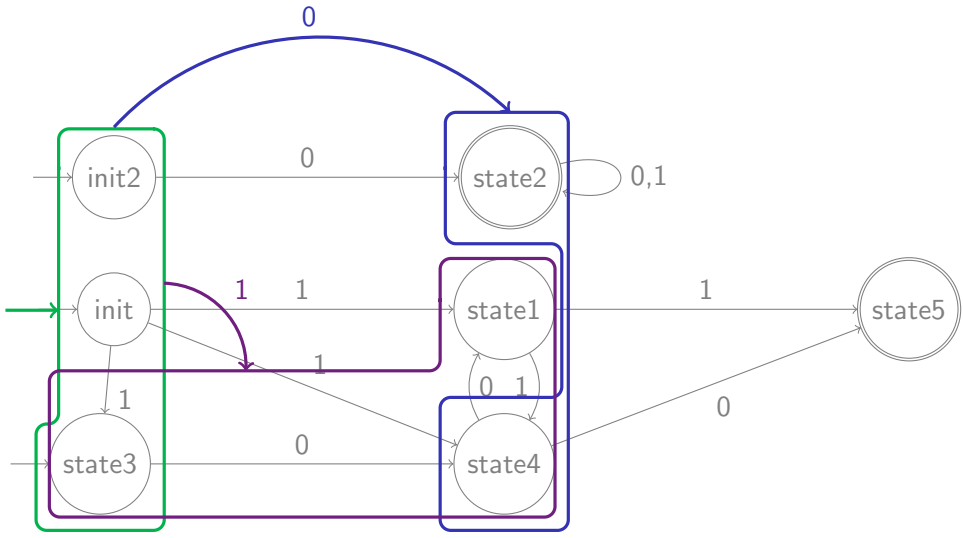
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